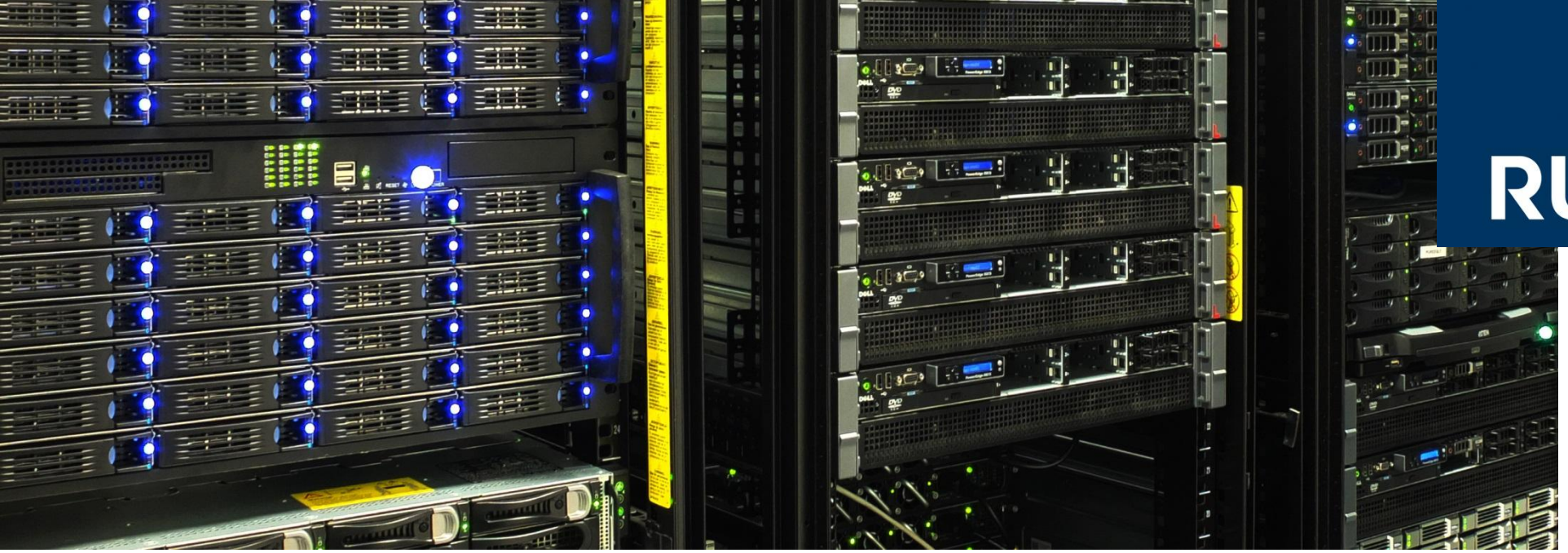


RUB

RUHR-UNIVERSITÄT BOCHUM

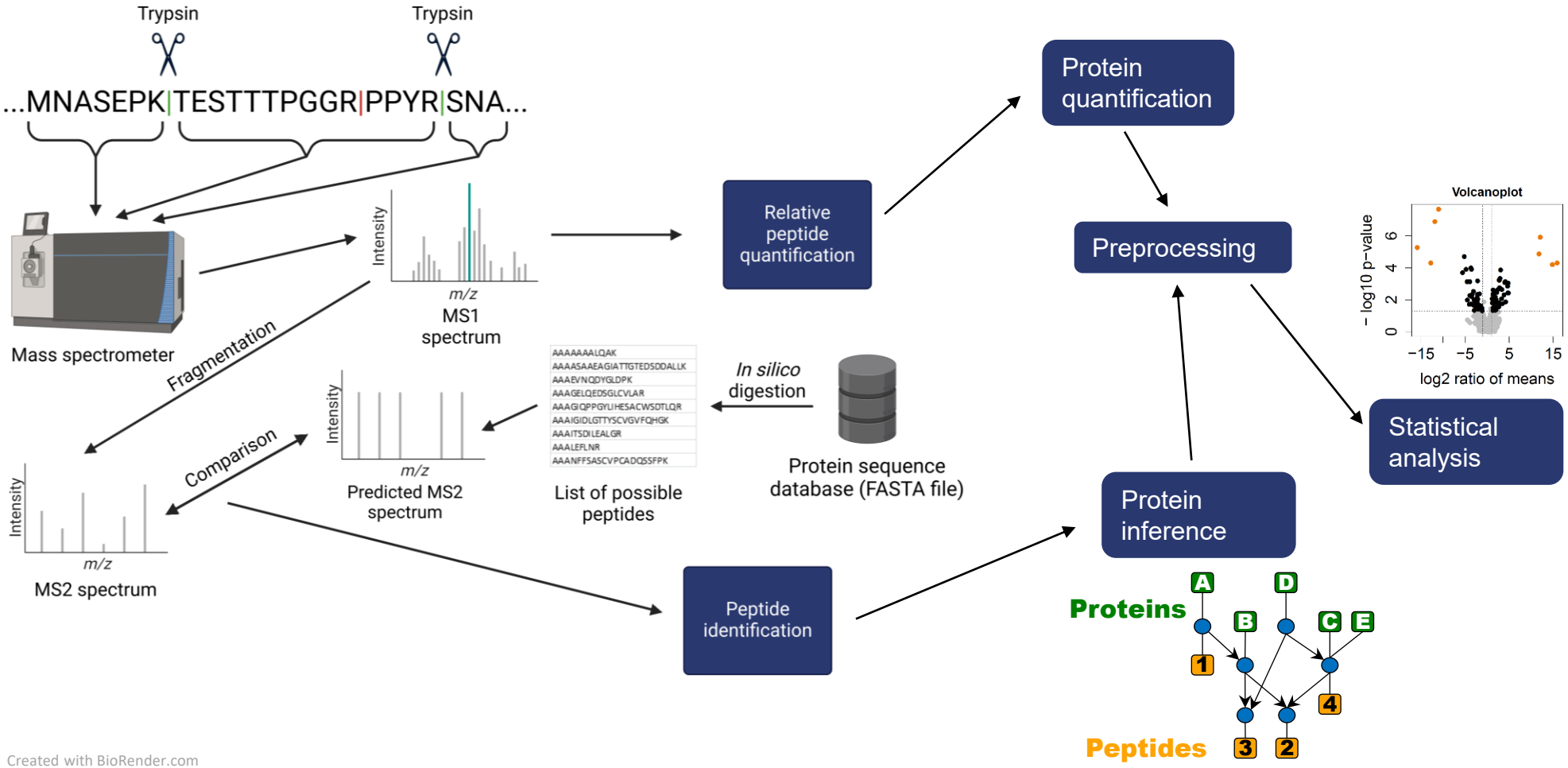
Bioinformatics for Biochemists

Tutorial Proteomics part II

Dr. Karin Schork
karin.schork@rub.de

Medizinische Fakultät
Medizinisches Proteom-Center
Medizinische Bioinformatik

Bottom-Up Proteomics Workflow

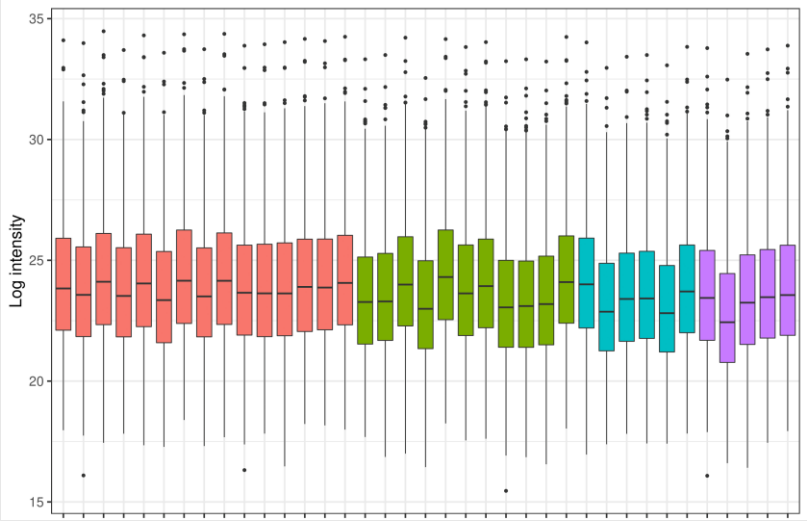


Contents

- Normalization and quality control of quantitative proteomics data
- Statistical tests (t-test)

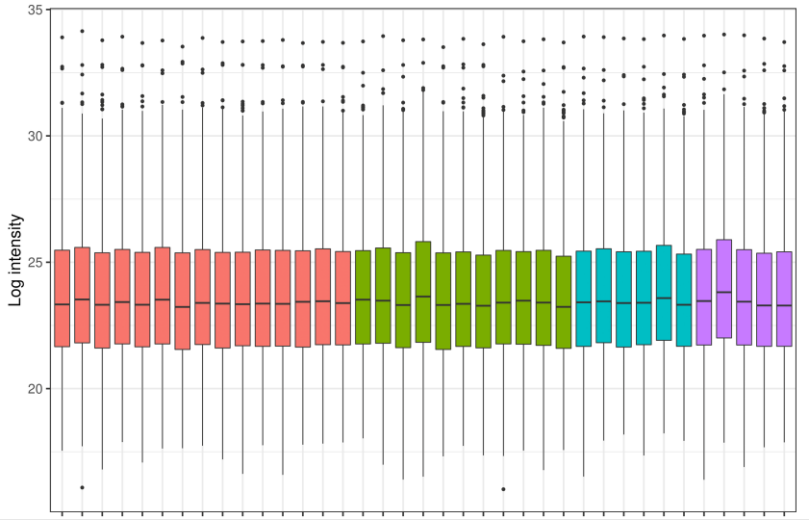
Normalization - Motivation

- Data contains technical and biological variation (but we are only interested in biological differences)
- **Reasons for technical bias (dt: „Verzerrung“):**
  - Small variations in experimental conditions and sample handling (temperature, age of column, pipetting)
  - Often exact reasons for bias are unknown



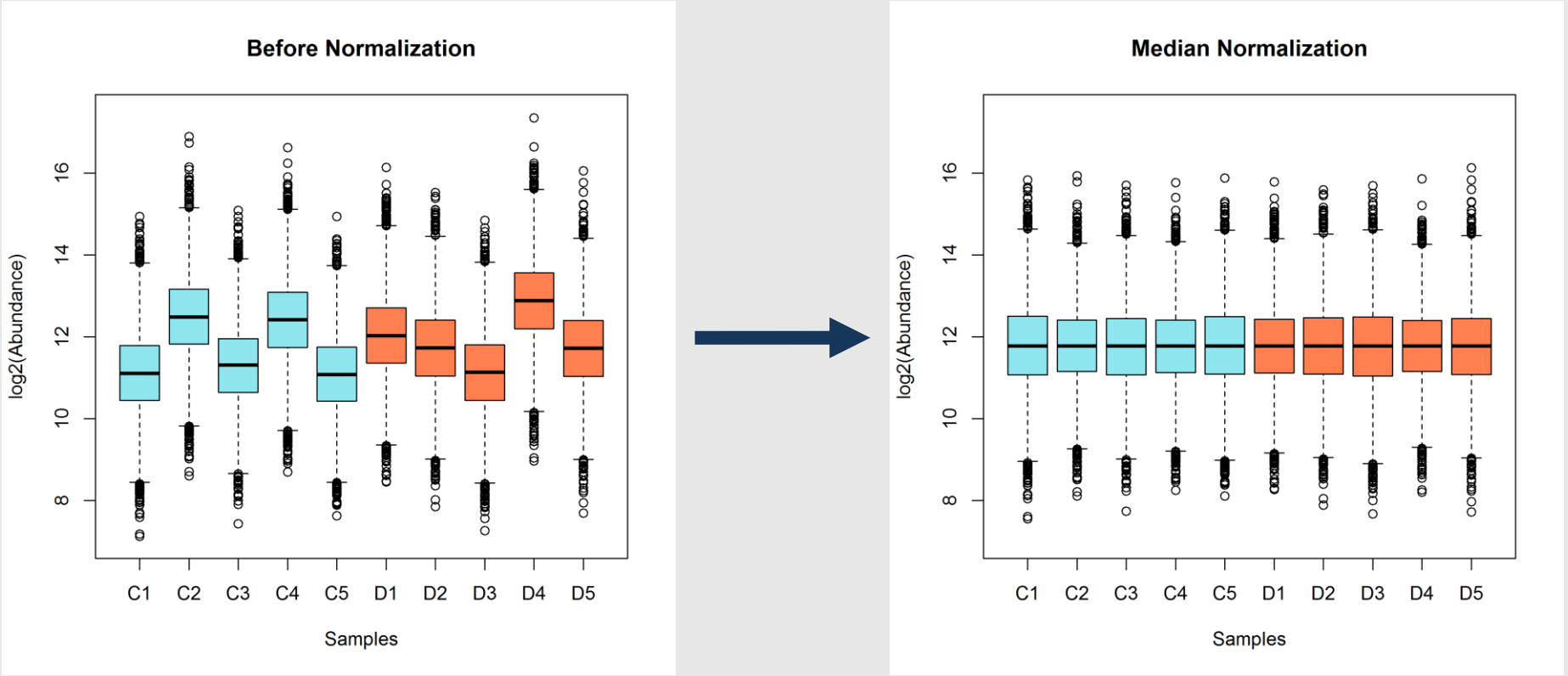
Normalization - Motivation

- Data contains technical and biological variation (but we are only interested in biological differences)
- **Aims of normalization:**
  - Reduce technical bias while keeping biological differences
  - Make samples better comparable
  - Make following statistical analysis more reliable



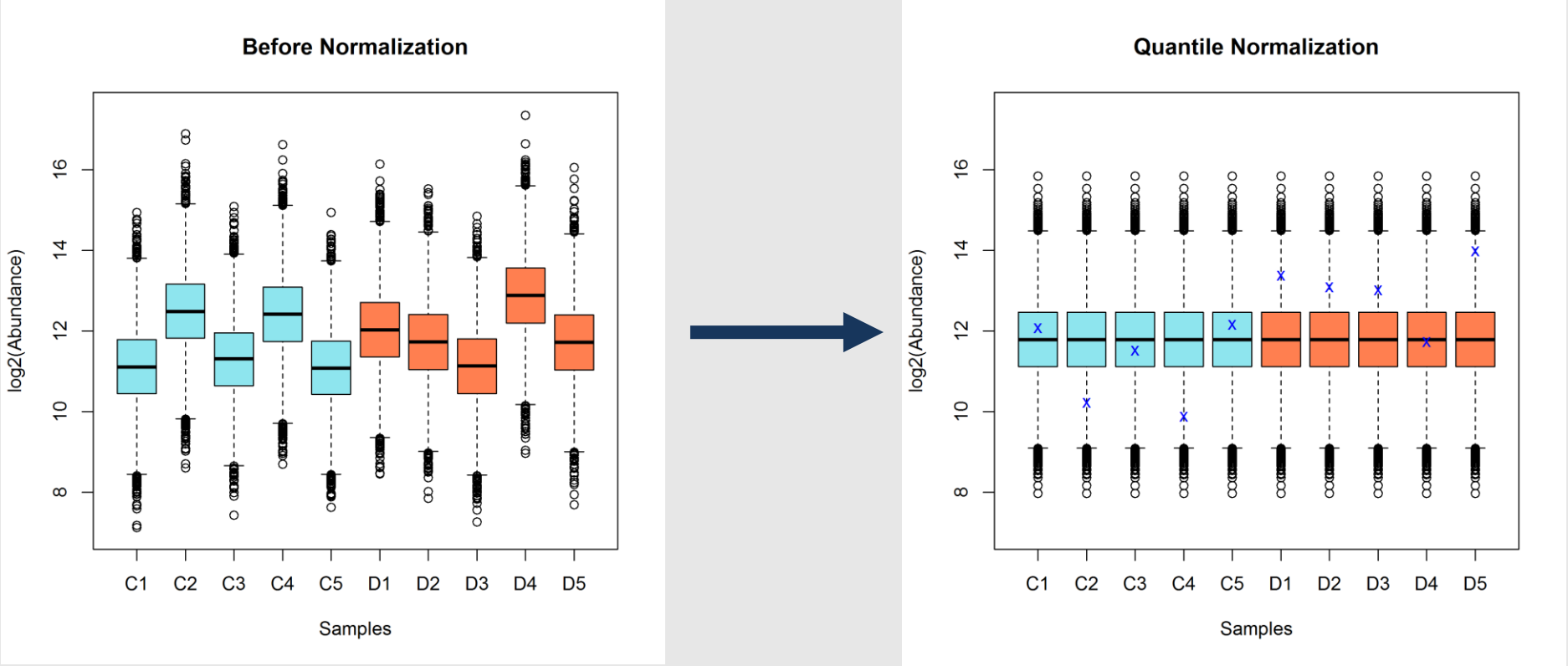
Median normalization

Shift or scale samples to have the same median



Quantile normalization

Normalize all samples to the same distribution



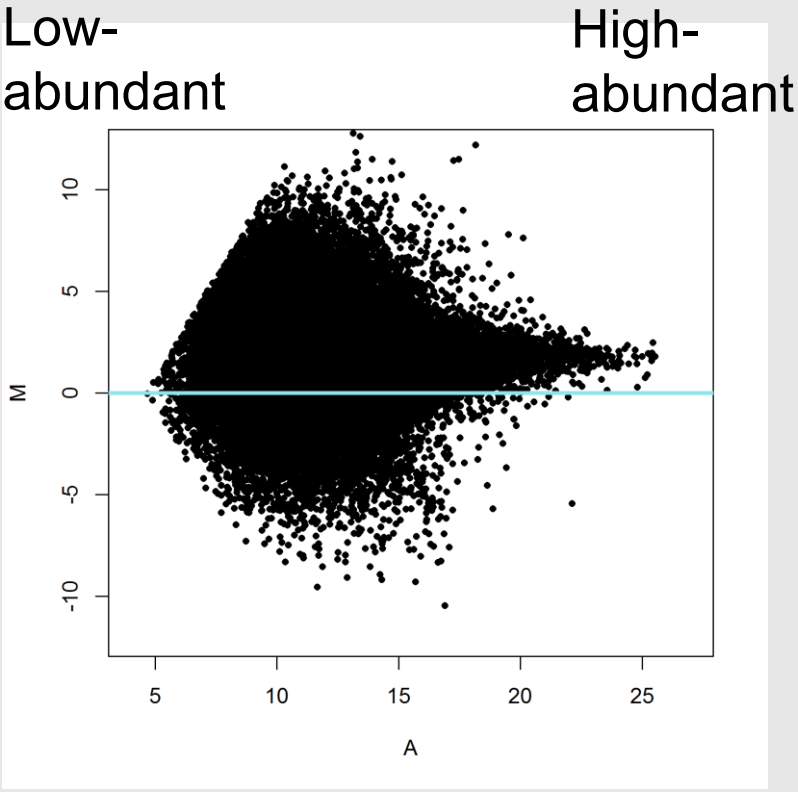
+ can normalize samples with different variances
- often too strict

LOESS-Normalization (LOcal regrESSion)

- Alternative name: LOWESS = Locally Weighted Scatterplot Smoothing
- Based on MA-Plots (compares 2 samples with each other)
- X-axis: average of log2-intensities (A: Average)
- Y-axis: difference of log2-intensities (M: Minus)
- Each point is one protein (or peptide)

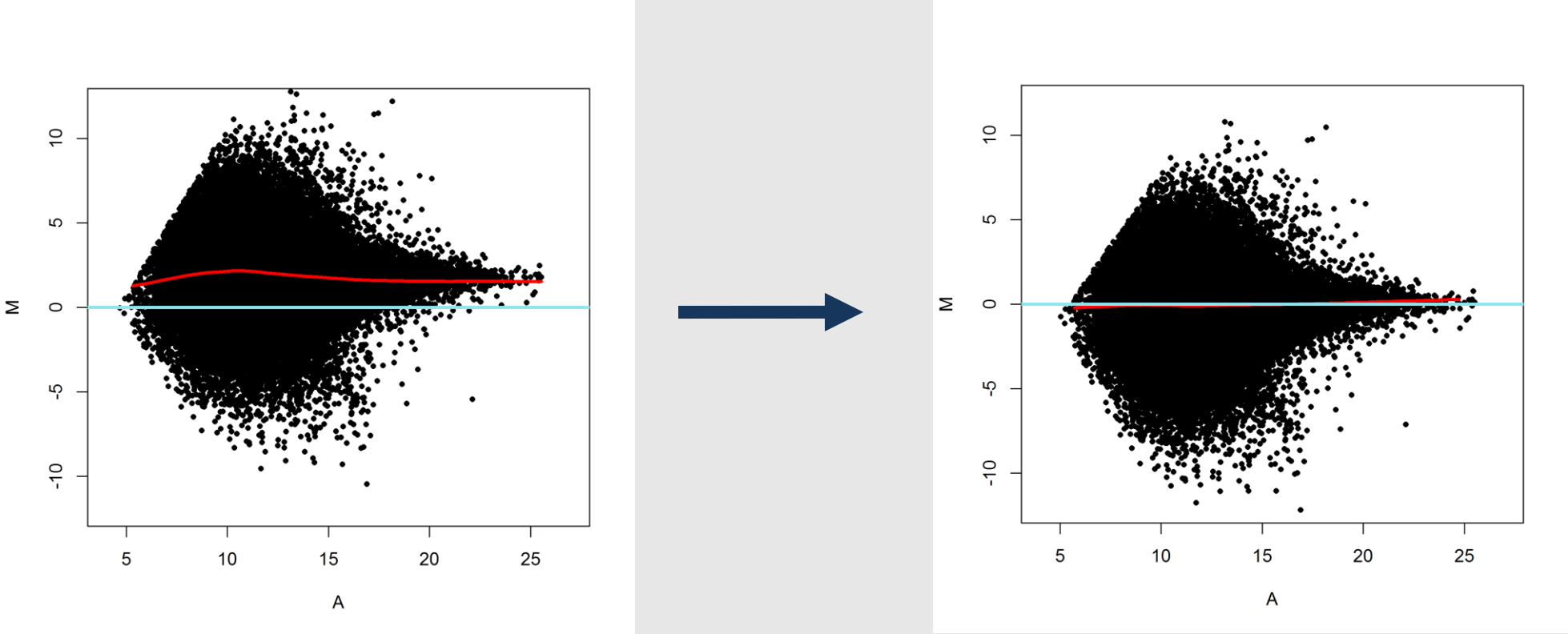
$$A = \frac{1}{2}(log_2(X_1) + log_2(X_2)) = \frac{1}{2} log_2(X_1 \cdot X_2)$$

$$M = log_2(X_1) - log_2(X_2) = log_2(X_1/X_2)$$



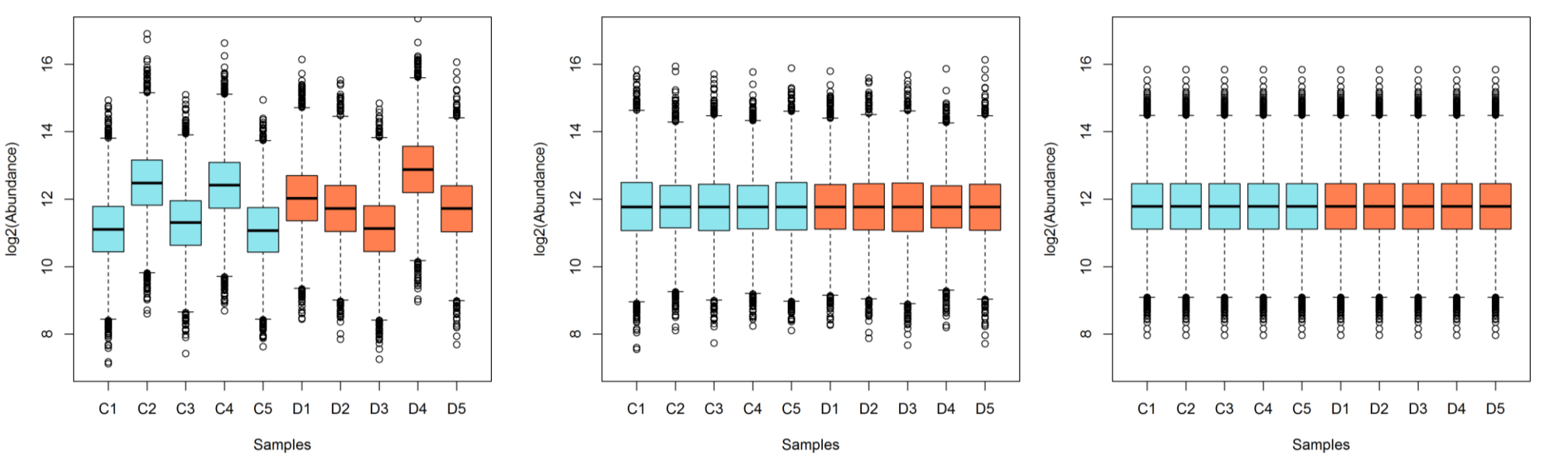
LOESS-Normalization

- Aim: symmetry of the scatterplot around M = 0 (blue line)
- Estimation of a local linear regression curve (red line) to remove bias
- Back-transformation of the normalized M and A values to intensities possible



Evaluation of normalization Boxplots

- Boxplots of can give a hint if normalization is needed
- Differences in variances may be visible (box sizes)
- Looking at boxplots alone is not enough to compare normalization methods! (e.g. all boxplots are always equal for quantile normalization)



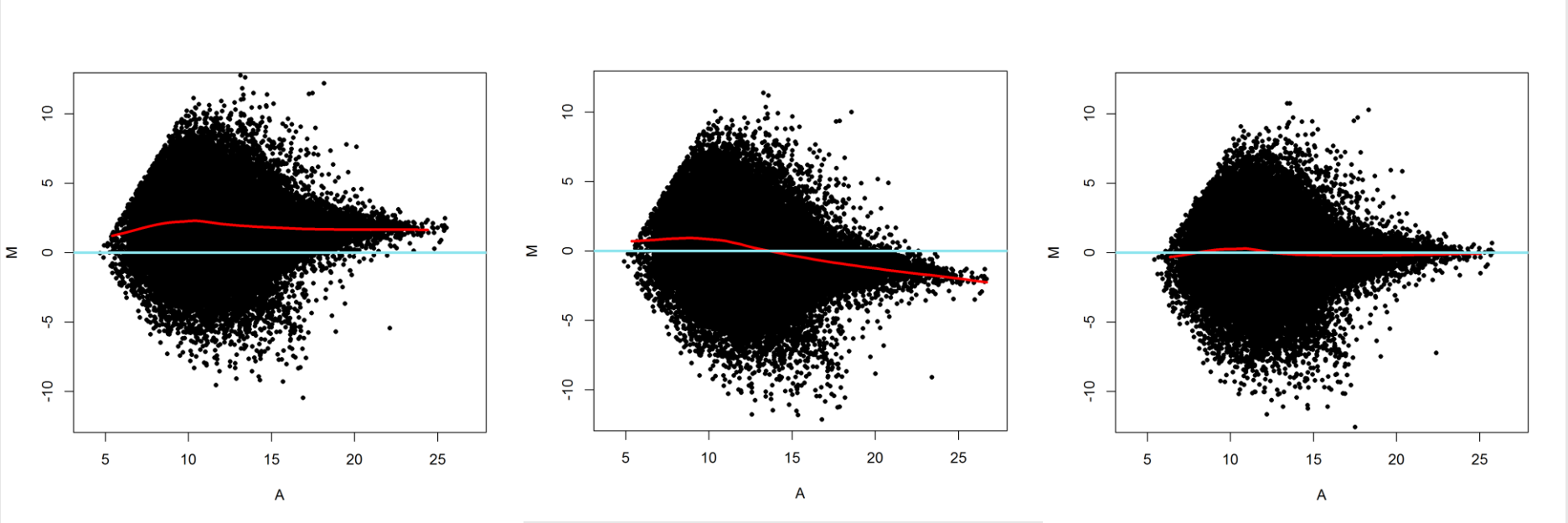






Evaluation of normalization MA-Plots

- important: look at all pairs of samples! → many MA-Plots
- Compare with unnormalized data



Unnormalized
→ bias ☹

→ unsuitable normalization for this dataset ☹

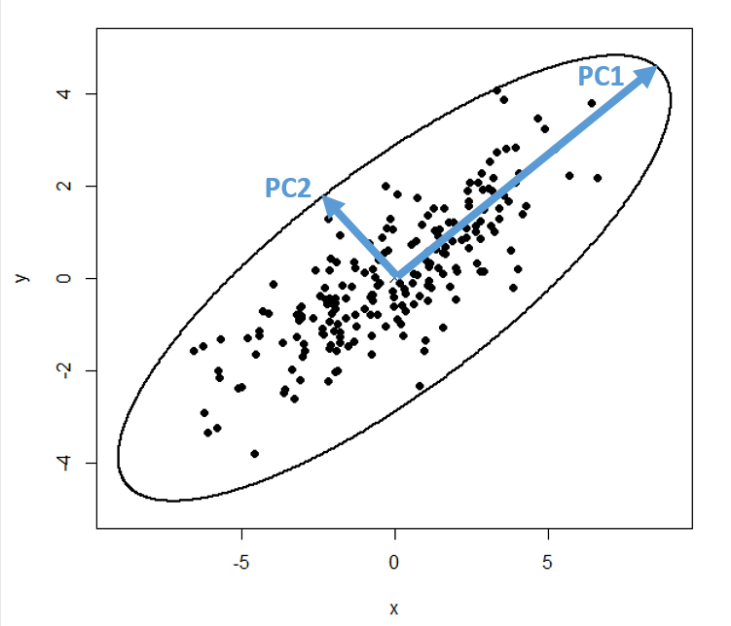
→ suitable normalization for this dataset ☺

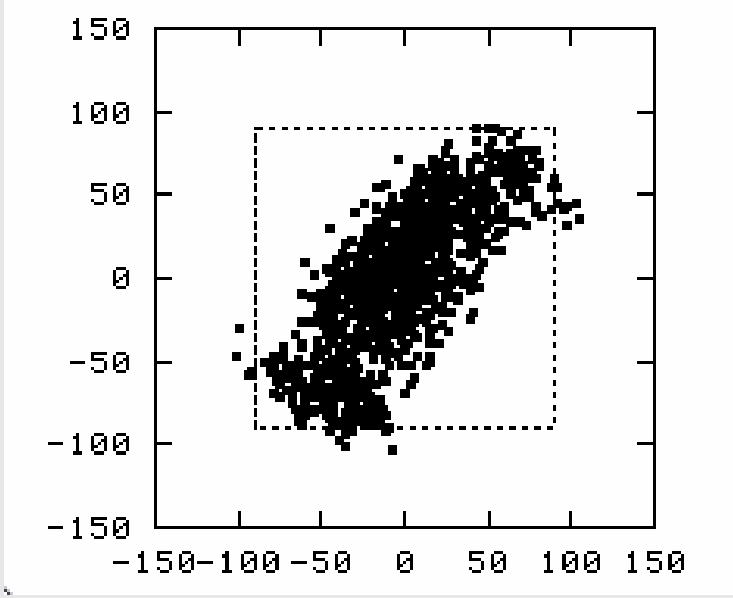
Principle Component Analysis (PCA)

- transform n variables $(X_1, X_2, \dots, X_n)$ into n Principal Components
- Transform the data into a new coordinate system (rotation)
- Principal Components (PCs) are linear combinations of the original variables:

$$PC = w_1X_1 + w_2X_2 + \dots + w_nX_n$$

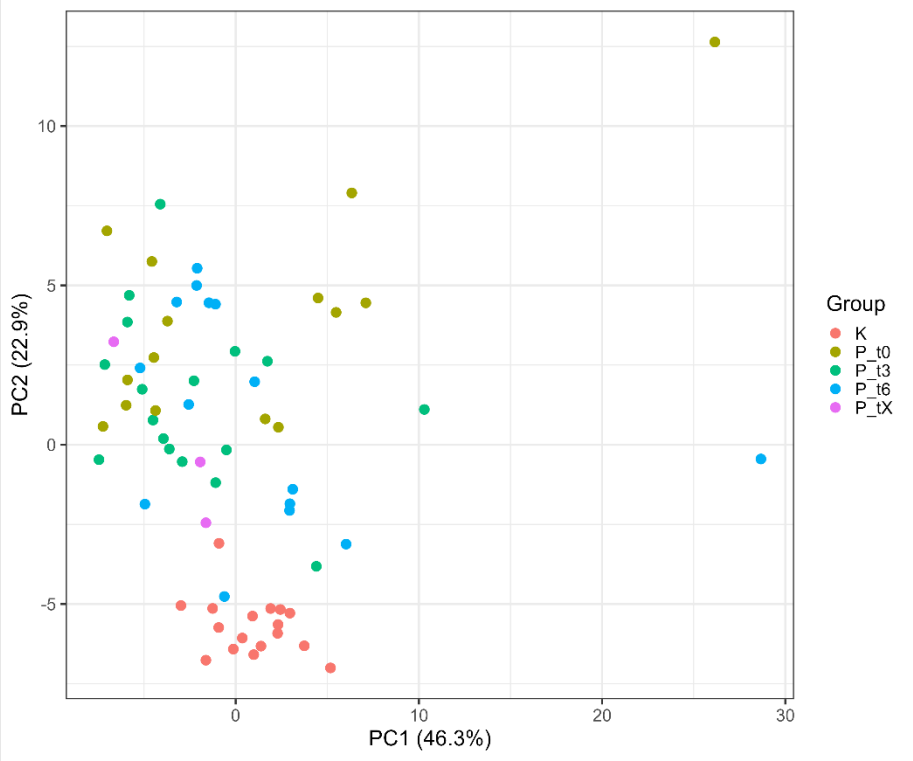
- PC1 has the largest possible variance, PC2 the largest variance and of vectors orthogonal to PC1, …
- 2-dimensional PCA-Plots as an overview over the whole data set

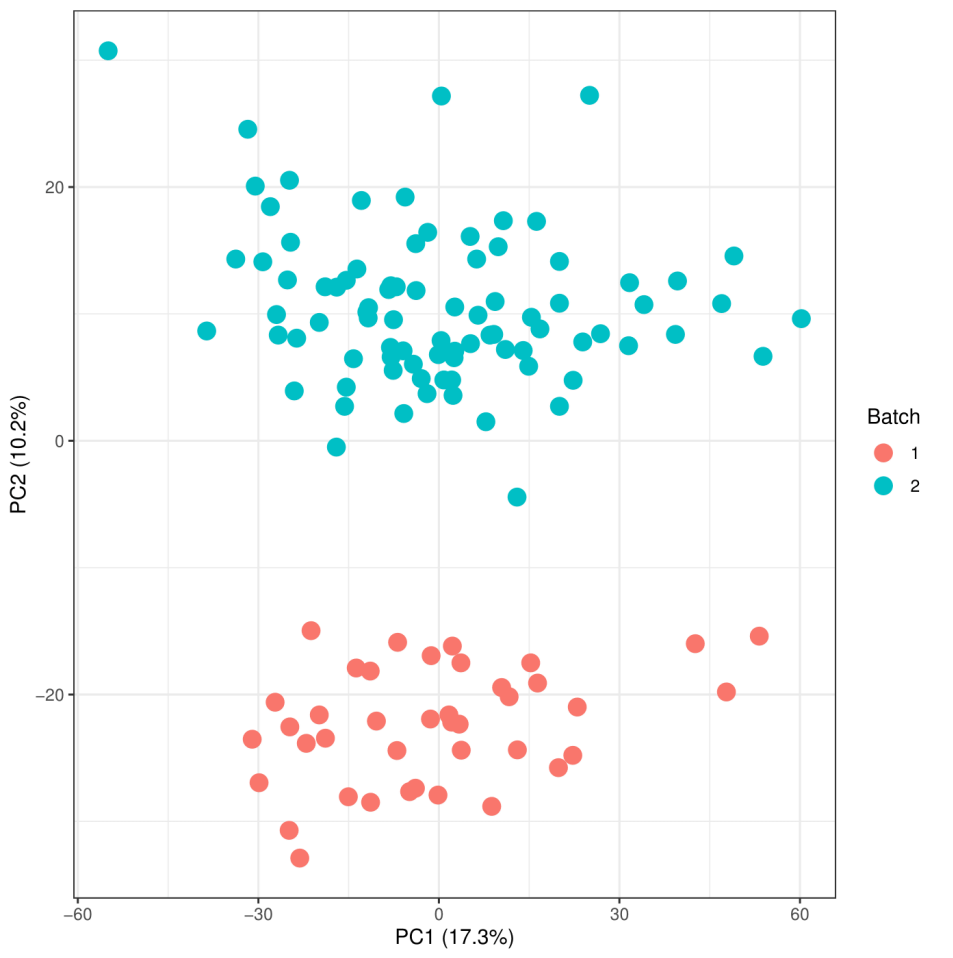




PCA in proteomics

- Overview over whole data set
- Detection of outlier points
- Detection of batch effects





t- test

- Tests for differences in location (means) of two groups
- n samples for group 1: $x_1, \dots, x_n$
- m samples for group 2: $y_1, \dots, y_m$

- Null hypothesis (H_0): There is no difference in the true means of group 1 and group 2:

$$H_0: \mu_x = \mu_y$$

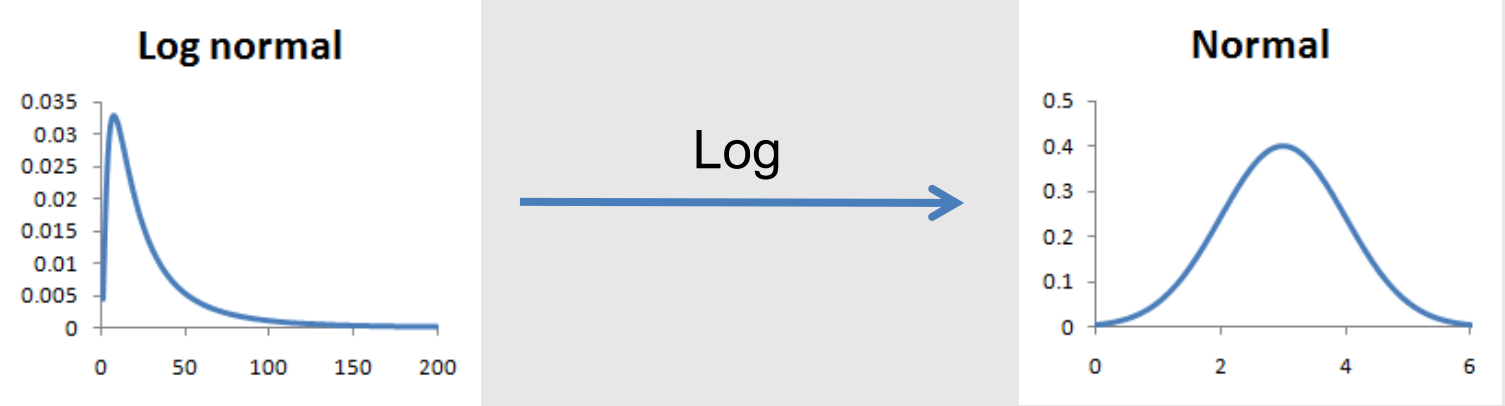
- Alternative hypothesis (H_1): There is a difference in the true means of group 1 and group 2:

$$H_1: \mu_x \neq \mu_y$$

- Decision to reject H_0 or not reject H_0 (not to accept H_0!)

Log-normal distribution

- The t-test assumes that the data follow a normal distribution
- Protein intensities are often not normal, but **log-normal distributed**
- That means that the log-transformed data follows a normal distribution
- Because of that, protein intensities are usually log-transformed before calculating the t-test



T-test: error types

- H_0: $\mu_x = \mu_y$ vs. H_1: $\mu_x \neq \mu_y$
- Error types:

Truth	Test decision	
	do not reject H_0	reject H_0
H_0 true	☺	☹ type I error (α)
H_0 false (H_1 true)	☹ type II error (β)	☺ power (1-β)

- Probability of Type I error (α) is controlled by the significance niveau (e.g. α = 5%)
- Probability of Type II error (β) cannot be controlled and depends on variance, sample size and difference of means

- Power: ability of the test to detect true changes of means

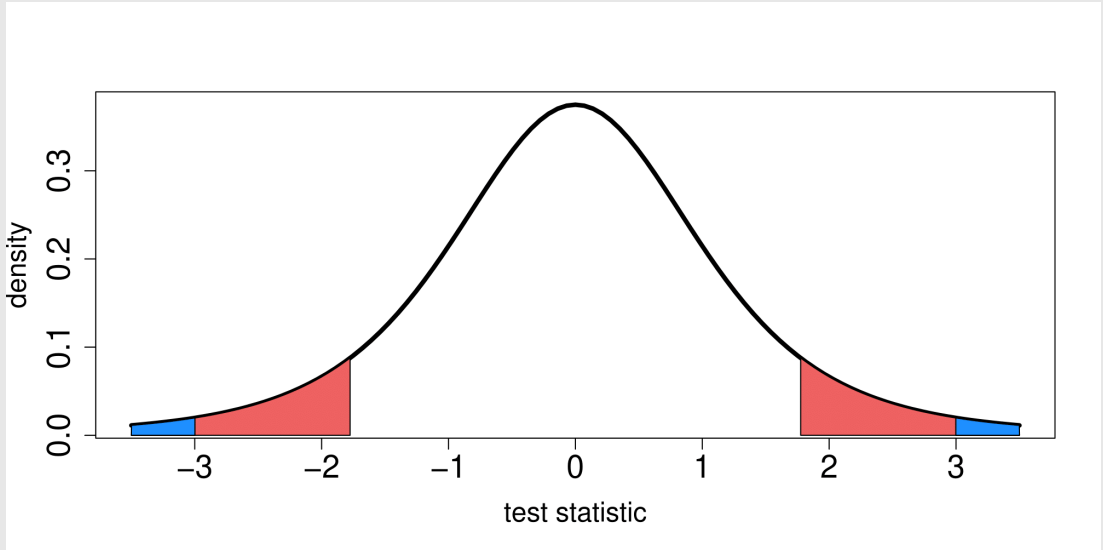
T-test: p-values

- T-test statistic (for the special case $m = n$, i.e. equal group sizes):

$$t = \frac{\delta}{\sqrt{\frac{1}{n}(s_X^2 + s_Y^2)}}$$

$\delta = \bar{x} - \bar{y}$: difference of mean values of the two groups
n: number of samples in each group
s_X^2, s_Y^2: sample variances in the two groups

- If H_0 is true, t follows a t-distribution
- Compare calculated t from your data to this distribution, is it extreme?
- P-value = propability (area) for a more extreme value
- If p-value < 0.05, it is considered significant



t= 1.77
→ p = 0.15

t = 3.00
→ p = 0.04

Paired t-test

Paired *t*-test

- Especially in medical applications paired designs are used
- Common examples: measurements on the same patients
  - pre/post treatment
  - tumerous/healthy tissue
- Common situation:
  - high variance between patients (inter-individual)
  - low differences of the measurements of the same patient (intra-individual)
- An unpaired t-test might not recognise existing differences between groups in this situation
- A paired t-test performs better because it only uses the intra-individual differences
- In **R**: `t.test(x, y, paired = TRUE)`

Fold change

- Measure for the effect size (biological relevance)
- Describes how much a quantity changes going from an initial value (A = mean of group 1) to a final value (B = mean of group 2)
- Definition: $FC = \frac{B}{A}$
- E.g., mean of protein intensity of the disease group divided by mean of control group (on original values before log-transformation)
- For analysis and visualization log fold changes are used (usually log2)
- FCs are usually calculated on the non-transformed protein intensities (in contrast to the p-values, for which usually the log-transformed intensities are used).

Examples:

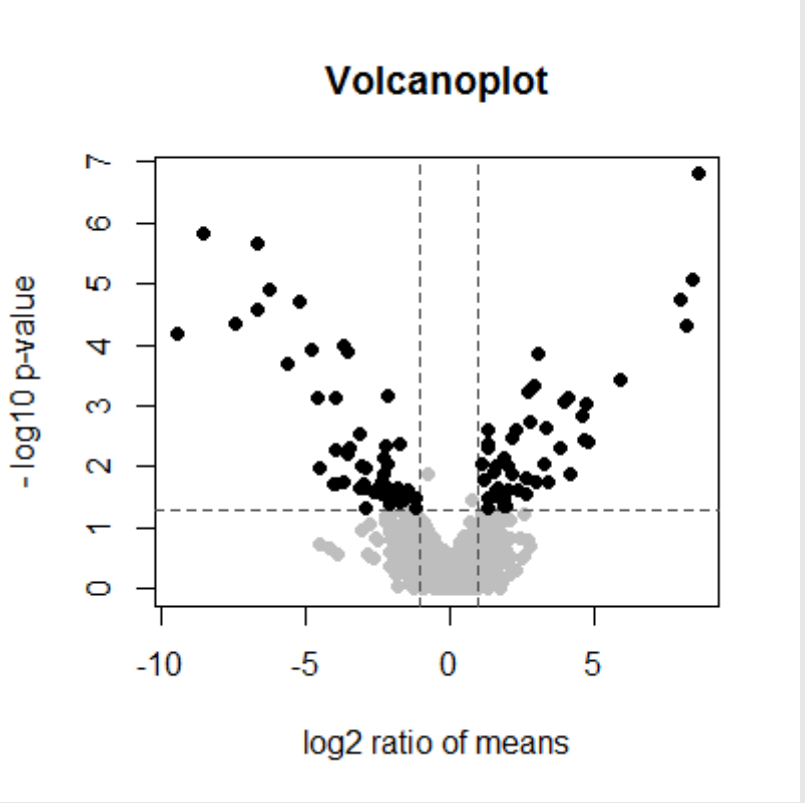
$$\mathrm{A} = 200{,}000 \text{ and } \mathrm{B} = 800{,}000 \Rightarrow FC = 4 \Rightarrow \log_2(FC) = 2$$

$$\mathrm{A} = 200{,}000 \text{ and } \mathrm{B} = 100{,}000 \Rightarrow FC = 0.5 \Rightarrow \log_2(FC) = -1$$

Candidate selection
Volcano plot

- For each protein, you have now calculated a p-value and a fold change
- Volcano plot shows $-\log_{10}(\mathrm{p}-\mathrm{value})$ vs. $\log_2(\mathrm{FC})$

- Promising biomarker candidates usually have a small p-value (≤ 0.05) and a large fold change (FC > 2 or < $1/2$)
- i.e. these proteins show up in the left and right upper corners of the volcano plot



File path navigation

- Function `getwd()` shows the current working directory
- `setwd()` changes your working directory
- **R** uses a forward slash '/' as a separator between folders (Windows normally uses backshlashes '\')
- **R** on Windows: use forward slashes or double backslashes
- Examples:

```
"C:/Users/Paul/Documents/Data/data.txt"
"C:\\Users\\Paul\\Documents\\Data\\data.txt"
file.path("Users","Paul","Documents","Data","data.txt") # construct relative path
```

ProtStatsWF

- R package actively developed at the MPC
- Contains workflows for statistical analysis of proteomics data

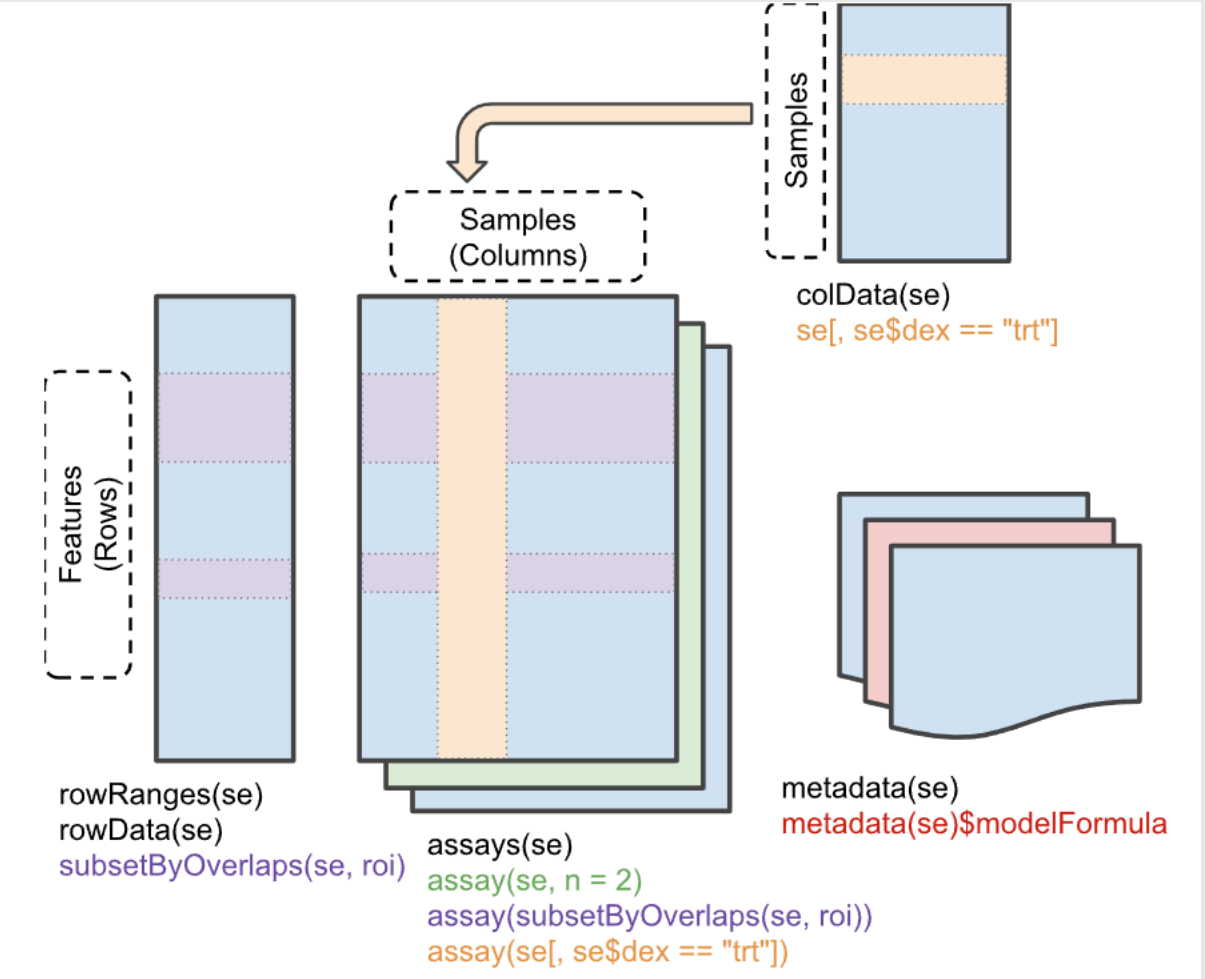
- prepareData() function
  - Imports proteomics + clinical data into a SummarizedExperiment object
  - Normalization
- workflow_QC() function
  - Valid value plot
  - Boxplots
  - PCA
  - MA-Plots
- ttest() function
  - Calculates t-test and fold changes
- VolcanoPlot_ttest() function
  - Volcano plot of t-test results

SummarizedExperiment (SE) object

- R object to store omics data with additional information on rows and columns
- Rows = proteins
- Columns = samples

- colData: contains additional clinical data (i.e. age, sex etc)
- prepareData() will give you a list of 2 objects:
  - SE
  - D_long
- You can access the SE object :

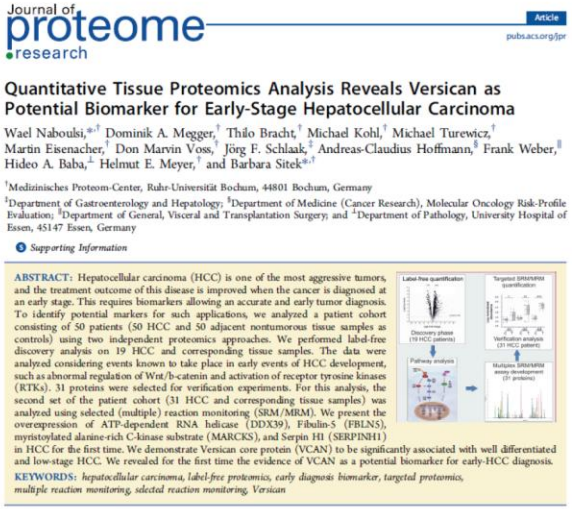
```
D <- prepareData(…)
D$SE
```



Today's data set

Dataset: HCC Proteomics

- `HCC_19vs19.xlsx` in the folder `Exercises/data`
- 19 samples of hepatocellular carcinoma tissue + adjacent non-tumorous tissue
- 19 vs. 19 paired design
- Quantitative proteomics data
- Results from differential analysis (p-values, fold changes)
- Additional table with further information on patients (age, gender, tumor staging, etc.): `clinical_data.xlsx`



Journal of proteome research — Article — pubs.acs.org/jpr

Quantitative Tissue Proteomics Analysis Reveals Versican as Potential Biomarker for Early-Stage Hepatocellular Carcinoma

Wael Naboulsi,*[,†] Dominik A. Megger,[†] Thilo Bracht,[†] Michael Kohl,[†] Michael Turewicz,[†] Martin Eisenacher,[†] Don Marvin Voss,[†] Jörg F. Schlaak,[‡] Andreas-Claudius Hoffmann,[§] Frank Weber,[∥] Hideo A. Baba,[⊥] Helmut E. Meyer,[†] and Barbara Sitek*[,†]

[†]Medizinisches Proteom-Center, Ruhr-Universität Bochum, 44801 Bochum, Germany
[‡]Department of Gastroenterology and Hepatology; [§]Department of Medicine (Cancer Research), Molecular Oncology Risk-Profile Evaluation; [∥]Department of General, Visceral and Transplantation Surgery; and [⊥]Department of Pathology, University Hospital of Essen, 45147 Essen, Germany

Supporting Information

ABSTRACT: Hepatocellular carcinoma (HCC) is one of the most aggressive tumors, and the treatment outcome of this disease is improved when the cancer is diagnosed at an early stage. This requires biomarkers allowing an accurate and early tumor diagnosis. To identify potential markers for such applications, we analyzed a patient cohort consisting of 50 patients (50 HCC and 50 adjacent nontumorous tissue samples as controls) using two independent proteomics approaches. We performed label-free discovery analysis on 19 HCC and corresponding tissue samples. The data were analyzed considering events known to take place in early events of HCC development, such as abnormal regulation of Wnt/b-catenin and activation of receptor tyrosine kinases (RTKs). 31 proteins were selected for verification experiments. For this analysis, the second set of the patient cohort (31 HCC and corresponding tissue samples) was analyzed using selected (multiple) reaction monitoring (SRM/MRM). We present the overexpression of ATP-dependent RNA helicase (DDX39), Fibulin-5 (FBLN5), myristoylated alanine-rich C-kinase substrate (MARCKS), and Serpin H1 (SERPINH1) in HCC for the first time. We demonstrate Versican core protein (VCAN) to be significantly associated with well differentiated and low-stage HCC. We revealed for the first time the evidence of VCAN as a potential biomarker for early-HCC diagnosis.

KEYWORDS: *hepatocellular carcinoma, label-free proteomics, early diagnosis biomarker, targeted proteomics, multiple reaction monitoring, selected reaction monitoring, Versican*